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ENGINEERING FACULTY
DEPARTMENT OF COMPUTER ENGINEERING

Operating Systems

First Assignment

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1. Installing the distribution

Abstract

In this report, I will talk about how I downloaded a virtual machine, Ubuntu, Git and Gcc. I will share my experiences and acknowledge with you about what my preferences was. I will also briefly explain what I know about Git, how to write a simple “hello world” program in C and its compilation with our simple Makefile.

a. Installing Virtual Machine

I had zero Linux experience, so I decided to download virtual machine because it is safer. I downloaded VirtualBox 6.1 for my Windows 10 host. I did not change anything while the set up. I also downloaded VirtualBox Extension Pack for incase of USB recognition needs.[1]

While creating a Virtual Machine I paid attention to give my name without spaces and special characters. I gave it a nickname “kastellan”. I selected the type as Linux and Version as Ubuntu(64-bit). In memory size page, I gave 2GB ram to my Virtual Machine after checking my ram usage at that moment from Task Manager’s performance page (I have 16GB ram maximum). In hard disk page I selected create a virtual disk now. In hard disk file type page, the page says, “If you do not need to use it with other virtualization software you can leave this setting unchanged”. So, I leave it as VDI (Virtual Disk Image). On the next page, storage on physical hard disk is asked. I preferred dynamically allocated and I highly suggest you too. It dynamically takes memory instead taking 10GB even you will not need that much space. As file location, I choose my “D:\” location because my “C:\” is full and I selected 10GB as maximum space. I suggest you choose “C:\” because it tends to work faster there. Our initial installation is over.[1]

Now we need to go to Settings on our Virtual Box. Then go to system tab and Processor tab. Most of our computers have multiple CPUs. In standard it uses 1 CPU. I have 16 CPUs, so I gave 4 CPUs which is more than enough. Do not give more than half of your CPUs! Now we press start our virtual machine. The virtual machine wants from us a operating system. We click add and choose .iso OS file.[1]

b. Installing Ubuntu

I preferred Ubuntu because it is a very popular, beginner friendly Linux distribution. I thought there would be more source because it is popular. Also, my goal was learning Linux, so I did not need a distribution like Kali that has hacking tools in it. I typed Ubuntu download on Google and I saw that the download .iso file is 2.67 GB. Because the download button on the page was downloading from my browser, I decided to find BitTorrent link (downloading as Torrent is lot faster and on browser downloading might fail if your connection is lost). Fortunately, Ubuntu's website support BitTorrent downloads too. By doing that I downloaded Ubuntu 20.04.02 LTS (long term support) version.

As language I choose English (because online sources are English). Then I selected install Ubuntu. I removed the tick on the "download updates while installing Ubuntu" option because If our set up is wrong it might end up with error. On next page, we choose to delete the disc and download Ubuntu (do not worry about warnings our disc is virtual and contain nothing). Later, we choose our country for time preference and our keyboard. Finally, we give username and password.[2]

After the setup we right click on desktop and select Terminal. In Linux we control our computer from Terminal by entering commands. I opened Terminal and typed "sudo su" which means admin says switch the user to admin, then I entered my password. In Linux we download from web, from Terminal too. I typed "apt-get update" which means update the downloader. I typed "apt-get upgrade" and performed the upgrading. Terminal wants confirmation and I type "y" for yes. Later, I typed "apt-get install" but I saw that it already did the installation, so I type exit twice. One for leaving the directory one for leaving the Terminal.[2]

2. Installing Git and setting up Github account

a. Installation and setting up

I opened Terminal and typed “sudo apt-get install git-core git-gui git-doc” and confirmed with pressing “y” as yes. After installation I typed “git --version” and got the output “gitversion 2.25.1” and be sure that Git is installed. I looked for a git bash but after for a quick search I learned that we could use Git from Terminal as well.[3]

For setup, I typed “git config --global user.name “cbarkinozer” to terminal and “git config --global “user.email”. This adds your sign to your changes so in a git team everyone can understand that you did the changes. After configuring these changes I had no output, so I thought it did not work but as soon as I typed “git config --list” I had the output of my configurations, so I be sure that it worked.[3]

b. Summary of Git

Git is a version control system that helps monitoring and synchronizing the code development process. Every user has a copy of their file on their computer and the server. You download the file on your hard drive then you do changes and send them to the server.[4]

There are four storing places for projects in Git. These are working directory, staging area, local repository(head) and remote repository(main). Working directory is the place you write your code. You Add your code to Staging area and commit it to local repository. If you are also the master of the repository (who controls the main branch) you can push it to main branch too. If you are not the master, then your code got reviewed and pulled to the main branch if it got approved. If there are different branches (it is suggested to not work on main branch), which there should, then the master might merge or fetch your branch with the main branch.[4]

If you wrote your code in working directory, there are three options. These options are staging modifying and committing. Staging means changed codes are not committed yet but marked to be committed to the local repository. Modifying means changed files are not stored in local repository yet so you can modify them. Committing means storing changed codes in the local repository.[4]

These are basic commands for Git [4]:

The command “git add” **adds** a file from **working directory** to **staging area**.

The command “git commit” **adds** all files from **staging area** to **local repo**.

The command “git push” **adds** committed files from **local repo** to **remote repo**.

The command “git fetch” **gets** files from the **remote repo** to the **local repo**.

The command “git merge” **gets** files from **local repo** to **working directory**.

The command “git pull” **gets** files from **remote repo** to **working directory**.

3. Installing Gcc and making a Makefile

a. Installing Gcc

I updated the packages by typing “sudo apt update” command on Terminal. I installed the build-essential package by typing “sudo apt install build-essential”. This command installs new packages named “gcc,” “g++” and “make”. I typed “gcc --version” and had output “gcc(Ubuntu 9.3.0-17ubuntu1~20.04) 9.3.0” and understood that gcc is ready to use.[5]

b. Writing a simple Makefile

First, I first created files on desktop one by one with the command “mkdir file_name”. I checked if it is in with “ls” command. I get in that file with “cd” command. I used git clone command to clone the files on my file with the command “git clone <https://github.com/CEN354/assignment1-cbarkinozer>”. I had to write and compile a program. I typed “nano hello.c” then I wrote a basic hello world C script. I pressed ctrl+x for quit. I entered the files name as “hello” and saved it[5]. Later, I created a text file named Makefile and typed the below Makefile code. [6]

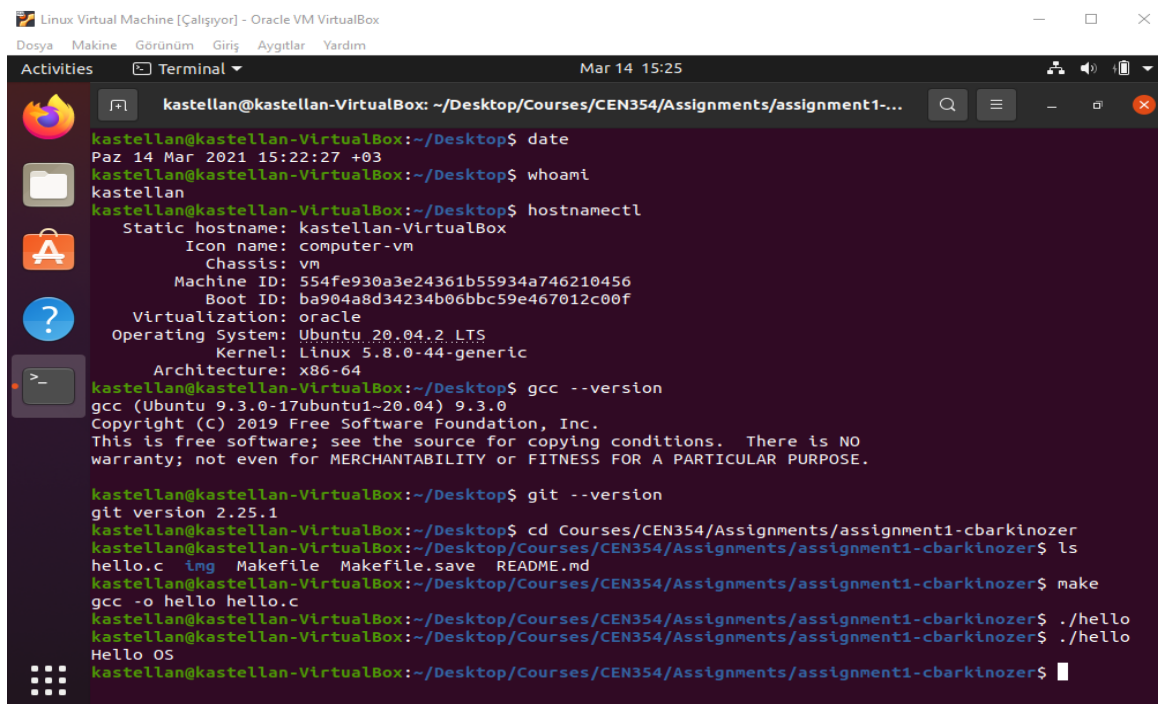
```
CC=gcc
SOURCES=hello.c
all: hello
hello: $(SOURCES)
    $(CC) -o hello $(SOURCES) #tab at the beginning
```

I had the error “Makefile:3: ***missing operator”. It is because using spaces instead of tab (The syntax must be exactly same). At the end I typed “make” command and had the output “gcc -o hello hello.c”.

Do not forget that if you type make command again it will “make: Nothing to be done for ‘all’”. You need to delete the hello file with “rm hello” command so we can have the “gcc -o hello hello.c” output. I set the drag drop option from devices tab to “work both sided” on virtual machine. Then I drag drop my Report.pdf to the file and took a screenshot of my terminal. First, I added the files that are not tracked with command “git add hello.c”. Later, I committed “git commit -m ‘The homework is done’” with commit message which is a must. At the end I pushed the final file with “git push origin master” command.

Summary

Virtual machines are good for experiencing a Linux operating system safely. I preferred Virtual Box application on my Windows 10 machine. I downloaded Ubuntu and explained my preferences. I preferred Ubuntu because it is a very popular, beginner friendly Linux distribution. Then I downloaded and configured Git. By doing that I learned how to install and use an application on Linux. Later, I briefly explained Git’s terms, commands, and the way of working. I downloaded Gcc to write a simple “hello world” program. I wrote a simple Makefile to compile my helloWorld.c file. At the end, I learned how to set up a Linux distribution on a virtual machine; download and set up Git and Gcc and, write and compile a code file. You can see on the bottom line that all working.



```
Linux Virtual Machine [Çalışıyor] - Oracle VM VirtualBox
Dosya Makine Görünüm Giriş Aygıtlar Yardım
Activities Terminal Mar 14 15:25
kastellan@kastellan-VirtualBox: ~/Desktop/Courses/CEN354/Assignments/assignment1-...
kastellan@kastellan-VirtualBox:~/Desktop$ date
Paz 14 Mar 2021 15:22:27 +03
kastellan@kastellan-VirtualBox:~/Desktop$ whoami
kastellan
kastellan@kastellan-VirtualBox:~/Desktop$ hostnamectl
Static hostname: kastellan-VirtualBox
Icon name: computer-vm
Chassis: vm
Machine ID: 554fe930a3e24361b55934a746210456
Boot ID: ba904a8d34234b06bbc59e467012c00f
Virtualization: oracle
Operating System: Ubuntu 20.04.2 LTS
Kernel: Linux 5.8.0-44-generic
Architecture: x86_64
kastellan@kastellan-VirtualBox:~/Desktop$ gcc --version
gcc (Ubuntu 9.3.0-17ubuntu1~20.04) 9.3.0
Copyright (C) 2019 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

kastellan@kastellan-VirtualBox:~/Desktop$ git --version
git version 2.25.1
kastellan@kastellan-VirtualBox:~/Desktop$ cd Courses/CEN354/Assignments/assignment1-cbarkinozer
kastellan@kastellan-VirtualBox:~/Desktop/Courses/CEN354/Assignments/assignment1-cbarkinozer$ ls
hello.c img Makefile Makefile.save README.md
kastellan@kastellan-VirtualBox:~/Desktop/Courses/CEN354/Assignments/assignment1-cbarkinozer$ make
gcc -o hello hello.c
kastellan@kastellan-VirtualBox:~/Desktop/Courses/CEN354/Assignments/assignment1-cbarkinozer$ ./hello
Hello OS
kastellan@kastellan-VirtualBox:~/Desktop/Courses/CEN354/Assignments/assignment1-cbarkinozer$
```

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